

Journal of Graduate Medical Education

The Utility of a Gap Year: Differences in academic productivity post dermatology residency --Manuscript Draft--

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Full Title:	The Utility of a Gap Year: Differences in academic productivity post dermatology residency
Short Title:	
Article Type:	Original Research
Keywords:	research year; gap year; medical education; residency application
Abstract:	Background Many students pursue a gap year to strengthen their applications to dermatology residency. While a dedicated research gap year has been reported to increase the chances of matching at top dermatology programs, it is unclear if this academic pursuit truly signals interest in or commitment to a career in academic medicine. Objectives This study investigated publication rates and academic appointments of NYU dermatology residency alumni in the 10 years following training to determine whether a gap year predicted higher long-term academic involvement. Methods Data was collected on NYU dermatology alumni with residency start dates from 1998 to 2014. Publicly available webpages were used to determine publication quantity, and authorship role in the 0-5 and 5-10 year windows following the completion of training as well as academic appointments at 5 and 10 years post training. Data was stratified by amount of time spent in medical school (4 years, 5-6 years, and 7+ years). Results There were no significant differences in the number of papers published per year in the 4 years vs 5-6 years groups 0-5, 5-10, or 0-10 years post-training. There were no significant differences in academic appointments between these two groups at 5 or 10 years post training. Conclusions This study suggests that completing a research year does not portend increased academic productivity. Admissions committees may want to reconsider the value they assign to the research year given the lack of evidence for differences in career-scale academic outcomes.



July 22nd, 2026

Gail M. Sullivan, MD, MPH
Editor in Chief
Journal of Graduate Medical Education

Dear Dr. Sullivan,

It is our pleasure to submit our original manuscript, "The Utility of a Gap Year: Differences in academic productivity post dermatology residency" for consideration for publication as a research article in *Journal of Graduate Medical Education*.

Many medical students applying to dermatology residency feel immense pressure to complete a research year to improve their chances of a successful match. Similarly, upon reviewing residency applications, many programs view the research year as a signal of academic interest, a desirable trait. Given the financial and temporal implications of pursuing a gap year, we feel strongly that academic advisors, students, and admission committees should have a thorough understanding of the implications and significance of the completion of a gap year. Although there are studies in dermatology literature that examine match rates after the completion of a gap year, to our knowledge, no studies have evaluated longitudinal differences in academic contributions in the years following dermatology training.

Our study examines the academic productivity and appointments of NYU dermatology residency alumni in the 10 years following the completion of their training. We compare rates of publications per year and academic appointments amongst those who completed gap years, additional advanced degrees, and individuals who completed 4-year medical degrees alone. We did not find statistically significant differences between those who completed a gap year and those who applied to dermatology residency with 4-year medical school degrees. Overall, our study highlights that the completion of a research year should not be used in isolation as a signal of academic commitment amongst those applying to dermatology residency. We appreciate your consideration and look forward to hearing back from you.

All authors have seen and approved the manuscript and contributed significantly to the work. The manuscript has not been previously published nor is being considered for publication elsewhere. There is no conflict of interest. Thank you very much for your time and consideration.

Sincerely,

A handwritten signature in black ink, appearing to read 'Miriam Keltz Pomeranz'.

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Abstract: 240 words

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4 **Abstract:**
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7 Background: Many students pursue a gap year to strengthen their applications to dermatology
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9 residency. While a dedicated research gap year has been reported to increase the chances of
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11 matching at top dermatology programs, it is unclear if this academic pursuit truly signals interest
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28 Methods: Data was collected on NYU dermatology alumni with residency start dates from 1998
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34 academic appointments at 5 and 10 years post training. Data was stratified by amount of time
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36 spent in medical school (4 years, 5-6 years, and 7+ years).
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43 Results: There were no significant differences in the number of papers published per year in the
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45 4 years vs 5-6 years groups 0-5, 5-10, or 0-10 years post-training. There were no significant
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47 differences in academic appointments between these two groups at 5 or 10 years post training.
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52 Conclusions: This study suggests that completing a research year does not portend increased
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54 academic productivity. Admissions committees may want to reconsider the value they assign to
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56 the research year given the lack of evidence for differences in career-scale academic outcomes.
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Introduction:

Applying to dermatology residency is a notoriously competitive and challenging undertaking. According to the NRMP, during the 2024 match cycle only 434 of the 740 senior applicants from allopathic medical schools and 30 of 101 senior applications from osteopathic medical schools successfully matched to dermatology, yielding a match rate of 58.6% and 29.7 for MD and DO seniors respectively [1]. Hopeful dermatology applicants feel immense pressure to participate in activities that strengthen their applications and improve their likelihood to match at programs of their choice. Participation in a gap year, when medical students add a year to their standard medical school curriculum to focus on research or other scholarly pursuit, is now frequently recommended by academic advisors and program directors to students planning to apply for dermatology residency. This recommendation is based on the perception that completing a gap year will improve an applicant's chances of a successful match. Previous studies have failed to find a significant relationship between research gap year completion and overall match rates within dermatology [2,3], but a research gap year may increase the likelihood of matching at top ranked institutions in dermatology [2].

The role of a research gap year from the position of the medical student is threefold. It provides students with focused time to generate additional research that can be listed on their applications, and more time for previous work to produce published manuscripts. Additionally, it allows students to gain experience in academic medicine and signals an interest in academia to program directors. Lastly, it provides a framework for applicants to develop relationships with members of the department they are working with, leading to mentorship opportunities and stronger letters

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4 of recommendation. Despite these benefits, the financial ramifications of completing a research
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6 year should not be underestimated. Financial considerations for research year students include a
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8 lost year of income, an additional year of living expenses, and for some, tuition fees.
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14 From the position of program directors and admission committees, it remains unclear how the
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16 completion of a research gap year should be interpreted during the application review process,
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18 particularly for programs that desire to match applicants that will remain in academia after the
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20 completion of their training. Although the value of this year has not been extensively studied in
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22 dermatology, a recent study indicated that the completion of a research year did not lead to
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24 differences in post-training publications, practice setting, or rate of obtaining fellowships for
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26 dermatology residents that achieved board certification in 2010. Only the obtainment of
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28 additional degrees was associated with increased publications, although research gap year and
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30 additional degree completion were both associated with higher H-indices [4]. The aim of this
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32 study was to utilize publicly available information about NYU dermatology residency alumni to
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34 determine if the completion of a research gap year in medical school correlates to increased
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36 academic productivity and involvement following the completion of residency or fellowship
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38 training.
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48 **Methods:**

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50 The study population included NYU dermatology alumni with residency start dates from 1998 to
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52 2014 from the NYU Department of Dermatology, excluding international medical school
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54 graduates. Duration of medical school and degrees were collected from ERAS applications.
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56 Information regarding the alumni's individual academic appointments and productivity
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4 following the completion of residency were collected from PubMed, Doximity, LinkedIn, or
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6 professional webpages. IRB exemption for this study was granted by the NYU IRB.
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11 The data collected included the number of years to obtain medical degree, other degrees obtained
12 during that time, number of publications and year of publication, authorship type (first, middle,
13 or last), fellowships completed, and academic appointments. Data was subdivided into time
14 windows of 0-5 years, 5-10 years, and 0-10 years following the completion of training, defined
15 as starting after June 30th of the final year of their residency or fellowship program. Academic
16 appointments at fixed time points of 5 or 10 years after the completion of training were recorded.
17 For the alumni that have not yet reached 10 years after the completion of their training, academic
18 appointment at the 10-year time point was considered ‘not applicable’. Since not all alumni have
19 reached 10 years post training, publications were analyzed as number of papers per year to
20 reduce bias.
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38 Data was stratified for the individual alumni by time spent in medical school (4 years, 5-6 years,
39 and 7+ years). Four-year and 5-6-year subgroups were compared in terms of authorship rates and
40 academic appointments, as seen in Table 1. All analyses were performed using R, and P-values
41 were calculated using the Kruskal-Wallis, Wilcoxon rank-sum or Fisher’s exact test. Paired
42 differences were also evaluated between yearly average number of papers, as seen in Table 2. P-
43 values for the paired year analyses were computed using a multiple comparisons test with
44 Scheffe adjustment. Data collection and analysis were completed in 2024.
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58 **Results:**

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4 In total, ninety-six residents were included. There was no statistically significant difference
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6 between the number of papers published per year in the 4 years (N=45) vs 5-6 years (N=31)
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8 groups amongst all time windows (0-5 years, 5-10 years, or 0-10 years post-training). Similarly,
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10 the difference between the number of first or last author papers published per year in the 4 years
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12 vs 5-6 years groups amongst all time windows (0-5 years, 5-10 years, or 0-10 years post-
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14 training) did not achieve statistical significance. In terms of academic appointments, there was
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16 no statistically significant difference between the 4 years vs 5-6 years groups at the time points
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18 of 5 years or 10 years post training. To detect a difference of 1 paper per year on average
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20 between the 4-years and 5-6 years groups (assuming a standard deviation of 2) with 80% power
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22 and 5% level of significance, 64 participants in each group would be needed.
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31 In the paired analysis of yearly averages, there was no significant difference in papers published
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33 per year in the 0-5 and 0-10 year windows amongst all pairings (4 years vs 5-6 years, 7+ years vs
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35 4 years, 7+years vs 5-6 years). A statistically significant difference was observed in the number
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37 of papers produced per year in the 7+ years vs 5-6 years students in the 5-10 year window;
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39 however, the difference in papers per year between the 4 years and 5-6 years groups did not
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41 achieve significance. There were no significant differences in first or last author paper
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43 publications per year amongst all pairings (4 years vs 5-6 years, 7+ years vs 4 years, 7+years vs
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45 5-6 years), in all time windows (0-10 years, 0-5 years, 5-10 years).
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53 Discussion:

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4 While a pilot study limited to one program, our study provides further information regarding the
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6 career-scale value of the gap year within the field of dermatology. These findings provide context
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8 for residency application review committees, suggesting that completion of the gap year should
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10 be cautiously interpreted. Certainly, the clinical and research experiences students gain during
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12 that time are of value, and personal impressions reflected in letters of recommendation from
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14 mentors during that year should be considered, but we advise against assuming sincere interest in
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16 an academic career based on the completion of an academic gap year alone.
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23 Favoring a gap year also brings up questions of equity, as taking additional time in medical
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25 training may be cost prohibitive. Beyond a lost year of income, debt may be accrued or
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27 prolonged with interest from an additional year of living expenses or, for some, added tuition
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29 fees. Even when research positions are funded, the average funding is just 250% of the federal
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31 poverty level, and 71.7% of gap year takers report financial burden [5]. Although it is impossible
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33 to discount the value of additional publications, personal connections, and research experience
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35 gained during a research year, the financial and temporal burden must be weighed against these
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37 benefits when advising students and considered when ranking applicants.
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45 Creating a diverse and representative workforce of dermatologists rests on the foundation of a
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47 fair residency application process [6]. This pursuit requires intense scrutiny of the most
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49 financially and temporally limiting factors, including the dedicated research year. The lack of
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51 evidence found for differences in career-scale academic outcomes in all categories studied
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53 (research productivity, academic appointment, and first/last authorship) implies that research
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55 year completion should not be over-interpreted during the residency application review process.
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Larger and more broadly representative studies are needed to further elucidate the value of the dermatology research year.

Conclusions:

Our study suggests that completing a research year does not portend increased academic productivity. Admissions committees may want to reconsider the value they assign to the research year given the lack of evidence for differences in career-scale academic outcomes.

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Table 1. Comparison of academic productivity between 4 year and 5-6 year MDs.

		4 Years (N=45)	5-6 Years (N=31)	P-value
Papers per year (years 0-10)	Mean (SD)	1.06 (1.56)	1.12 (1.97)	0.84
	Median [Min, Max]	0.40 [0, 8.3]	0.40 [0, 10.7]	
Papers per year (years 0-5)	Mean (SD)	1.30 (1.33)	2.01 (3.56)	0.566
	Median [Min, Max]	0.80 [0, 5.2]	0.80 [0, 19.2]	
Papers per year (years 5-10)	Mean (SD)	0.80 (2.42)	0.15 (0.25)	0.718
	Median [Min, Max]	0 [0, 14.6]	0 [0, 1.0]	
First/last author papers per year (years 0-10)	Mean (SD)	0.58 (1.28)	0.63 (1.55)	0.644
	Median [Min, Max]	0.20 [0, 7.5]	0.20 [0, 8.3]	
First/last author papers per year (years 0-5)	Mean (SD)	0.63 (0.75)	1.14 (2.78)	0.590
	Median [Min, Max]	0.40 [0, 3.6]	0.20 [0, 15.0]	
First/last author papers per year (years 5-10)	Mean (SD)	0.53 (2.14)	0.08 (0.19)	0.820
	Median [Min, Max]	0 [0, 13.2]	0 [0, 0.8]	
Academic appointment 5 years out of training?	No	29 (64.4%)	15 (48.4%)	0.239
	Yes	16 (35.6%)	15 (48.4%)	
	Missing	0 (0%)	1 (3.2%)	
Academic appointment 10 years out of training (if applicable)?	No	16 (35.6%)	9 (29.0%)	0.757
	Yes	10 (22.2%)	7 (22.6%)	
	Missing	19 (42.2%)	15 (48.4%)	

Footnote. Subgroup Analysis – P-values calculated using Wilcoxon rank-sum test or fisher's exact test

Table 2. Differences amongst the mean of the variables between paired years comparing academic productivity between 4 year, 5-6 year, and 7+ year MDs.

		Difference	Lower CI	Upper CI	P-value
Papers per year (years 0-10)	5-6 Years - 4 Years	0.062	0.062	1.026	0.988
	7+ Years - 4 Years	0.750	-0.360	1.861	0.279
	7+ Years - 5-6 Years	0.689	-0.496	1.874	0.387
Papers per year (years 0-5)	5-6 Years - 4 Years	0.715	-0.618	2.048	0.445
	7+ Years - 4 Years	0.622	-0.913	2.157	0.629
	7+ Years - 5-6 Years	-0.093	-1.731	1.545	0.629
Papers per year (years 5-10)	5-6 Years - 4 Years	-0.656	-1.698	0.386	0.329
	7+ Years - 4 Years	0.896	-0.304	2.095	0.211
	7+ Years - 5-6 Years	1.552	0.271	2.832	0.019
First/last author papers per year (years 0-10)	5-6 Years - 4 Years	0.051	-0.692	0.795	0.987
	7+ Years - 4 Years	0.392	-0.464	1.248	0.554
	7+ Years - 5-6 Years	0.341	-0.573	1.255	0.675
First/last author papers per year (years 0-5)	5-6 Years - 4 Years	0.509	-0.453	1.470	0.455

	7+ Years - 4 Years	0.343	-0.764	1.450	0.762
	7+ Years - 5- 6 Years	-0.165	-1.347	1.016	0.946
First/last author papers per year (years 5-10)	5-6 Years - 4 Years	-0.456	-1.341	0.429	0.474
	7+ Years - 4 Years	0.447	-0.573	1.466	0.582
	7+ Years - 5- 6 Years	0.903	-0.185	1.990	0.148

Footnote. P-values are computed using multiple comparisons test with Scheffe adjustment.